

思考题

$$2. \Phi = kA\Delta T \quad k = \frac{1}{\frac{1}{h_1} + \sum_{i=1}^n \frac{\delta_i}{\lambda_i} + \frac{1}{h_2}}$$

传热系数与表面传热系数 h 有关，与导热层的厚度，导热系数 λ ，与流体的性质，流体的流动，和热边界条件都有关。

$$7. \Phi = kA(t_{f1} - t_{f2}) = kA\Delta t_m$$
$$\Phi = q_{m1} c_{p1} (t_{i1}' - t_{i1}'') = q_{m2} c_{p2} (t_{i2}' - t_{i2}'')$$

12-3

$$t_f = 80^\circ\text{C} \quad \rho = 971.8 \text{ kg/m}^3 \quad c_p = 4.18 \text{ kJ/kg}\cdot\text{K} \quad \lambda = 67.4 \times 10^{-2} \text{ W/m}\cdot\text{K}$$
$$v = 0.365 \times 10^{-6} \text{ m}^2/\text{s} \quad \eta_f = 358.1 \times 10^{-6} \text{ kg/m}\cdot\text{s}$$

$$Pr = 2.21$$

$$Re = \frac{u_{\infty} d}{v} = \frac{5(80 \times 10^{-3})}{0.365 \times 10^{-6}} = 1095890$$

$$\text{用湍流 } Nu_f = 0.027 Re_f^{0.8} Pr_f^{1/3} = 2387.6725$$

$$h = Nu \frac{\lambda}{d} = 20116.1409 \text{ W/m}^2\cdot\text{K}$$

$$\text{假设 } t_{w1} = 79.98^\circ\text{C}$$

$$q = \pi d h \Delta t = 101.1148 \text{ W/m}$$

$$q = \frac{\Delta t}{\frac{1}{2\pi\lambda} \ln \frac{d_2}{d_1}}$$

$$t_{w2} = t_{w1} - q \times \frac{1}{2\pi\lambda} \ln \frac{d_2}{d_1} = 79.2^\circ\text{C}$$

$$t_m = \frac{t_{w1} + t_{w2}}{2} = \frac{79.98 + 79.2}{2} = 79.6^\circ\text{C}$$

$$C_p = 1.005 \text{ kJ/kg}\cdot\text{K} \quad \lambda = 2.805 \times 10^{-2} \text{ W/m}\cdot\text{K}$$

$$v = 17.597 \times 10^{-6} \text{ m}^2/\text{s} \quad \alpha = \frac{1}{T_m} = \frac{1}{273.15 + 41.435} = 0.003129 \text{ K}^{-1}$$

$$\mu = 19.4218 \text{ kg/m}\cdot\text{s}$$

$$Pr = 0.6984$$

$$Gr Pr = \frac{g \alpha \Delta T d^3}{\nu^2} \cdot Pr = \frac{9.81(0.003129)(79.2 - 20)(0.08)^3}{(17.597 \times 10^{-6})^2} (0.6984)$$

$$= 1.8 \times 10^6 \Rightarrow C = 0.59 \quad n = 1/4$$

$$Nu_f = C(Gr \cdot Pr)^n = 0.59 (1.8 \times 10^6)^{1/4} = 19.8$$

$$h = \frac{\lambda}{d} Nu = 7.003 \text{ W/m}^2\cdot\text{K}$$

$$q' = h A (t_w - t_f) = \pi d h (79.2 - 20) = \underline{100.6847 \text{ W/m}}$$

$$\underline{t_w = 79.2^\circ\text{C}}$$

12-8

$$1) \dot{Q} = \dot{m}_2 c_{p2} (t_{v1}' - t_{v1}'') = -3(4.174) (20 - 50) = 375.66 \text{ kW}$$

$$t_{vm} = \frac{t_{v1}' + t_{v1}''}{2} = \frac{20 + 50}{2} = 35^\circ\text{C}$$

$$\Delta t' = t_{i1}' - t_{v1}'' \\ = 50^\circ\text{C}$$

$$2) \dot{Q} = \dot{m}_1 c_{p1} (t_{i1}' - t_{i1}'') = 375.66 \text{ kW}$$

$$\Delta t'' = t_{i1}'' - t_{v1}' \\ = 40^\circ\text{C}$$

$$\dot{m}_1 = \frac{375.66}{2.1(40^\circ\text{C})} = 4.4721 \text{ kg/s}$$

$$3) \dot{Q} = k A \Delta t_m \quad \Delta t_m = \frac{\Delta t' - \Delta t''}{\ln\left(\frac{\Delta t'}{\Delta t''}\right)} = 44.81^\circ\text{C}$$

$$k = 300 \text{ W/m}^2\text{K}$$

$$A = \frac{\dot{Q}}{\Delta t_m \cdot k} = 27.9447 \text{ m}^2$$